Data Science Life Cycle in Netflix

Introduction

Netflix is a popular online movie streaming app. To keep users happy, Netflix uses a **Data Science Life Cycle** to recommend the best movies and TV shows to each user. This document explains the 6 main steps of this process.

Step 1: Data Collection :

Netflix collects data every second from its users to understand what they like to watch. There are three main types of data collected:

- **Explicit Data (Direct Actions):** This includes things that the user intentionally does on the app.
  - Clicking **Thumbs Up / Thumbs Down** on a movie.
  - Typing keywords in the **Search Bar**.
  - Adding movies or series to **"My List"**.
- **Implicit Data (Hidden Actions):** This is user behavior data captured automatically in the background.
  - **Watch Duration:** How many minutes you watch a specific show.
  - **Skip Patterns:** When you press pause, fast-forward, or rewind.
  - **Context:** What time you watch (day vs. night), what day it is (weekday vs. weekend), and what device you use (Mobile, Laptop, or Smart TV).
- **Content Metadata:** Information about the movies, such as genres (Action, Horror, Comedy), names of actors, directors, and release year.

Step 2: Data Cleaning :

Raw data collected from servers has many errors, duplicates, and unwanted items. We must clean it so that our machine learning models work correctly.

- **Removing Accidental Clicks (Outliers):** If a user clicks a movie by mistake and exits the video within 10 seconds, it is dirty data. The cleaning system deletes these short logs so they do not affect recommendations.
- **Filling Missing Values:** If a user's device type or location is missing due to a network glitch, the system looks at that user’s past history and fills the blank space with their most frequently used device (like "Mobile").
- **Removing Duplicates:** Sometimes, due to app crashes, the same watch action is recorded 3 or 4 times. Python scripts like drop_duplicates() are used to remove extra rows and keep only one unique log.
- **Data Normalization:** A 20-minute episode and a 120-minute movie cannot be compared directly. The system converts absolute watch time into percentages (between 0.0 and 1.0) so the computer can understand it uniformly.

Step 3: Exploratory Data Analysis - EDA :

Before making predictions, data scientists use simple graphs and charts to look for interesting patterns in the cleaned data.

- **Time Analysis:** Line charts show that kids' cartoons are watched mostly on Saturday mornings, while thriller movies peak on Friday and Saturday nights.
- **Device Analysis:** Bar charts help identify that users watch short series on mobile phones during travel, but watch high-quality movies on Smart TVs at night.
- **Popularity Trends:** Pie charts show which genres are trending in a specific state or country (e.g., Tamil movies trending in Tamil Nadu).

Step 4: Data Modeling (Machine Learning Stage):

This is the heart of the project where algorithms are used to predict what a user wants to watch next.

- **User Similarity (Collaborative Filtering):** If User A and User B watch the same 10 movies, the computer understands their tastes are identical. If User A watches a new 11th movie and likes it, the system automatically recommends it to User B.
- **Tag Matching (Content-Based):** If you watch a lot of science-fiction movies, the system tracks the "Sci-Fi" tag and recommends other movies with the same tag.
- **Dynamic Poster Images:** Changing the poster image of a movie based on user taste. If you like comedy, Netflix will show you a funny poster for a movie. If you like romance, it will show you a romantic poster for the exact same movie!

Step 5: Deployment (Live Processing):

Once the machine learning model is ready, it must be put on live servers so that millions of users can see recommendations instantly on their screens.

- **Offline Layer:** Heavy data calculations run once a day or every night on large cloud databases (like Amazon Web Services - AWS) to create a basic list of recommendations for you.
- **Online Layer (APIs):** When you open your mobile app, lightweight microservices run instantly. They take your latest click or search query and update your homepage layout in less than a second using safe API connections.

Step 6: Maintenance & Monitoring (Sustain Stage)

User interests change over time, so a model cannot remain the same forever. It requires continuous maintenance.

- **Tracking Trend Shifts (Drift):** User habits change drastically during specific times (e.g., everyone searches for Christmas movies in December). The system monitors these shifts and updates the models.
- **A/B Testing:** Before giving a new update to everyone, Netflix gives it to only 5% of users (Treatment Group) and compares them with the other 95% (Control Group). If the 5% group spends more time watching videos, the new model is safely rolled out to all users.
- **Automated Retraining:** The system is set up with automated loops that pull fresh data every week to retrain the models, ensuring that newly released movies are instantly recommended to the right audience.